

2 GOING GREEN WITH EV CHARGING

The automobile sector has seen a huge increase in the development of vehicles that run on clean fuels. In India, the growth of electric vehicles (EVs) is on the rise. Yet, one of the main reasons why people willing to invest in a cleaner option for mobility are holding back is range anxiety arising from lack of enough charging stations.

To overcome this hurdle, startup Meras Plugins was established in 2017 to develop advanced charging solutions and public charging infrastructure across India. Founded by Vivek Swamynathan, Ragavendra Ravichandra, Satheesh Venkatachalam, and Raghavendra TS, it is currently incubated with IIT Madras Research Park (RTBI).

Ragavendra Ravichandran, co-founder, Meras Plugins, says, "We started with an aim to promote a shift towards e-mobility, which helps in reducing carbon footprints. The company was running as a bootstrap startup until the development of prototypes. The fully developed prototypes were tested at IIT Madras battery centre for various parameters including a smart feedback communication system. Our chargers, which have been sold to Amaron batteries, Academy of EV technology, have shown promising results in comparison to other standard chargers."

The company offers charge stations that enable easy charging on the go. The chargers operate on power ranging from 3.3kW to 50kW and can be lever-

aged for residential, corporate, and industrial purposes. AC slow charging stations developed by the company enable charging of three EVs at a time. The AC chargers operate on three-phase input and deliver power ranging from 3.3kW to 14.5kW. The AC Type 2 chargers range from 7kW to 22kW.

In DC fast-charging stations, a single module provides an output of 5kW (24V 200A), which can be paralleled with four other units at a time to achieve 25kW. These fast chargers are integrated with multiple communication protocols (CHAdeMO, CCS, and GB/T), which makes them compatible with all types of EVs.

These are compliant with Bharath standard (AC001 and DC001) and equipped with a complete protection function to ensure robustness, reliability, and security. Additional features include mobile app integration for connecting to the cloud and allowing users to spot the nearest charge stations, Wi-Fi or GSM-based cloud interaction, RFID-based user authentication, and hassle-free payments.

The startup has also built an IoT-based, scalable automotive telematics platform with the latest cloud-based technologies to gain valuable information (such as vehicle tracking, fuel metrics, and driver efficiency) and analytics in real time. The platform is capable of handling millions of events per second and can be used to optimise the overall cost in the system.

Ravichandran adds, "The optimal-



ly-built algorithm schedules charging for maximum utilisation of grid resources. Our flexible control system provides us with a network that can be connected to various systems and devices for information exchange. Energy management software and mobile app will further help people to adopt EVs."

When it comes to future plans, Ravichandran says, "In April 2020, we participated in Open Charge Alliance plugfest along with leaders in the EV sector globally to test the efficiency of our charging station and charging management system. Besides supporting EV manufacturers through delivering high quality, algorithm-driven chargers, we also aim to be a significant contributor in transforming the EV sector by setting up comprehensive charging infrastructure with our future proof cloud solutions. Being the first company to test OCPP 2.0.1 from India, the company is currently focusing on further international expansion."

—Ayushee Sharma

3 SELF-DRIVING ELECTRIC VEHICLE FOR DELIVERY

While growth of the Internet and its access to the masses has made it possible for e-commerce businesses to start in different parts of the world, it is the reliable delivery service that has been the backbone of these flourishing businesses today. From food and medicines to clothes and other consumer essentials, items can be ordered from

anywhere via mobile phones. With lockdowns imposed in various nations due to the Covid-19 pandemic, the need has grown further.

Nuro, a 2016 startup founded by Dave Ferguson and Jiajun Zhu, aims to utilise technologies like robotics and artificial intelligence (AI) for common well-being. The team has designed, prototyped, and extensively tested a

fully autonomous electric vehicle (EV) to make transportation of local goods more accessible, including groceries, prescription drugs, dry cleaning, and much more.

For self-driving, the vehicle uses its onboard sensors (like high-definition cameras, radar, ultrasonic sensors, and audio sensors) to provide a 360-degree live image of the surrounding